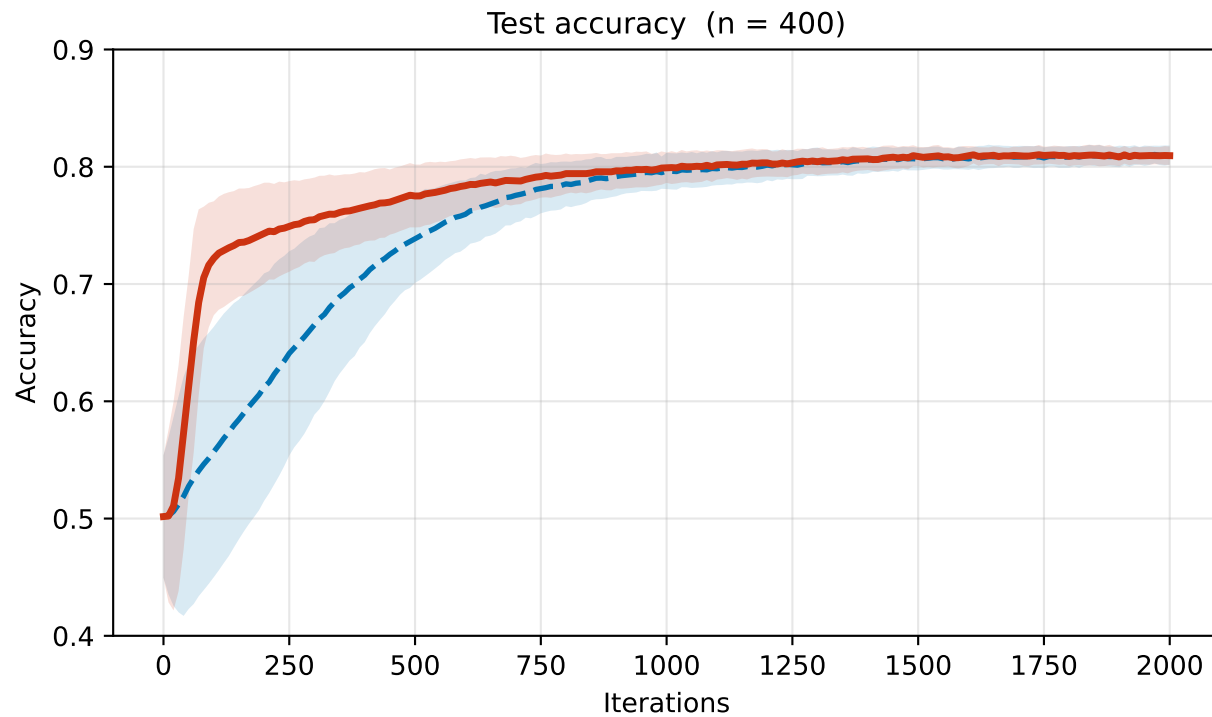
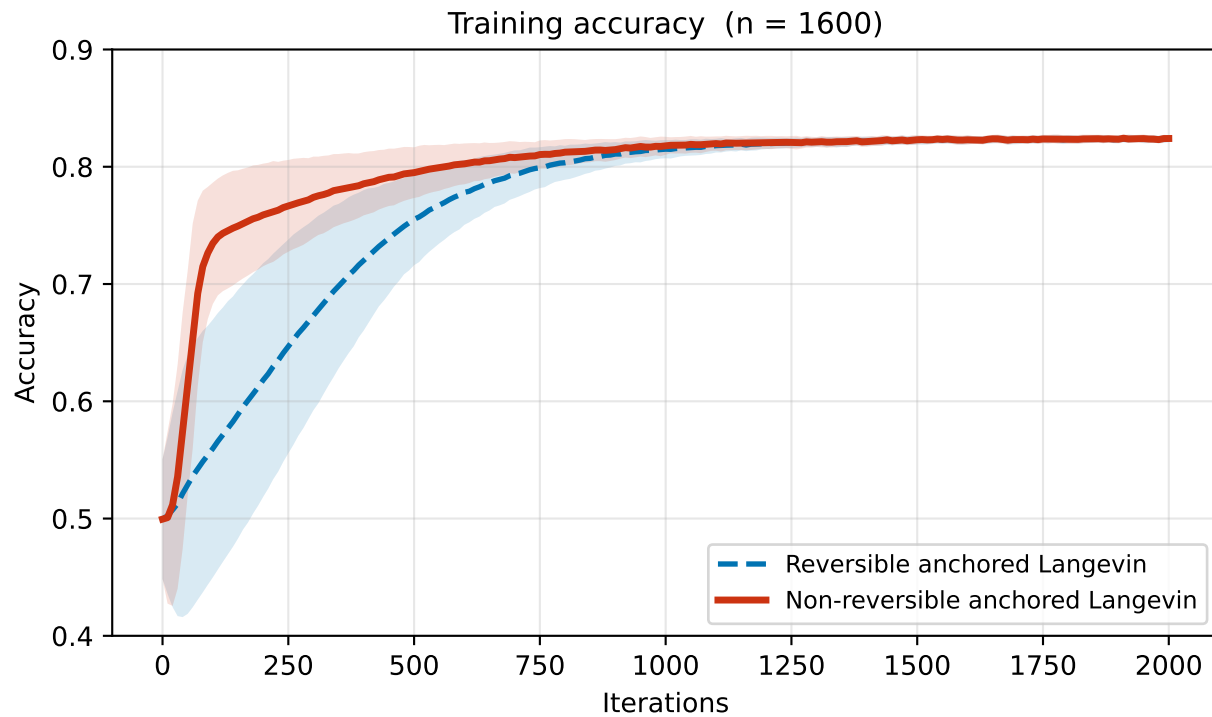


Non-reversible beats reversible anchored Langevin — L1-smooth ball



d = 9 (intercept + 8 slopes); constraint: L1-smooth ball; U = f + g with g = 2*sum|w_j| (non-differentiable), anchor delta = 0.02; EXACT gradient; eta = 7.0e-06; s = 4; R = 100; alpha = 0 vs 1.

Design: unstandardised covariates -- in each slope triple the covariance is 1 q1q1' + 64 q2q2' + 2 q3q3' with q1 = (1,1,1)/sqrt3 (the rotation axis), q2 = (0,1,-1)/sqrt2 (fast, no signal), q3 = (2,-1,-1)/sqrt6 (slow, in the rotated plane, carries signal); beta_triple = (1.5, 0.25, 0.25).

Lines: across-replicate mean; bands: mean +/- 1 sample sd of single-iterate accuracy (repeat-run variability, not confidence intervals). Shared initialisation and noise per replicate. y-axis fixed to (0.4, 0.9) -- NOT windowed on the plateau.